

Wenyi Wang

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RESEARCH INTERESTS

My core research interests are at the intersection of computer systems and machine learning, centered on building fast, scalable, and efficient solutions by exploiting extremely fine-grained parallelism. My current projects focus on large-scale, efficient LLM inference on many-node HPC systems, alongside the advancement of fine-grained parallelism techniques in OpenMP tasking models.

EDUCATION

University of Chicago

Ph.D. Computer Science, GPA: 4.0/4.0, Advisors: [Kyle Chard](#), [Ian Foster](#)

Sept. 2022 — Jun. 2028

Chicago, IL

Northwestern University

M.S. Computer Science, GPA: 3.9/4.0, Advisor: [Peter Dinda](#)

Sept. 2019 — Mar. 2021

Evanston, IL

Northeastern University

B.E. Software Engineering, Major GPA: 3.9/4.0, Advisor: [Tao Ren](#)

Sept. 2015 — Jun. 2019

Shenyang, China

PROJECTS

MLSys

ExaServe: Large-Scale LLM Serving on Exascale HPC Systems

- Built ExaServe, a pip-installable framework that turns a declarative YAML specification into a reproducible, large-scale LLM serving deployment on HPC systems, handling batch-scheduler integration, MPI launch, accelerator selection, and node-local model-weight staging.
- Deployed vLLM serving on ALCF Aurora from 1 to 256 nodes (3,072 vLLM replicas); non-streaming inference scales nearly linearly to 27.1k requests/s (3.8M tokens/s).
- Identified barriers to exascale LLM serving: a centralized streaming proxy that plateaus at about 4.7k requests/s and an $O(N^2)$ Ray Serve control-plane bottleneck (about 30 minutes of cluster bring-up at 256 nodes).

MPI-vLLM: Large Scale LLM Serving

- Enabling fast model-switching capabilities with MPI backend on limited GPU resources.
- Designing scalable workflows to deploy MPI-vLLM on hundred-nodes HPC systems.
- Conducting comprehensive experiments to study MPI-vLLM's scalability and performance

HPC

GNU-XTask: Optimizing Fine-Grained Parallelism Through Dynamic Load Balancing on Multi-socket Many-core Systems

- Enabled extremely fine-grained parallelism leveraging lock-less data structures XQueue on GNU-OpenMP.
- XQueue-enabled GNU-OpenMP shows up to 1522.8× performance improvement on an eight-socket machine with 192 threads.
- Developed GNU-XTask that is equipped with NUMA-aware, lock-less dynamic load balancing techniques that can bring 4× additional performance improvement over XQueue-enabled GNU-OpenMP

UDSHMEM: High-performance SHMEM Library for UpDown Accelerators (Advised by [Yanqing Li](#) and [Andrew A. Chien](#))

- Designed and implemented UDSHMEM data movement library based on SHMEM's specification on top of the novel UpDown computer system, written in UpDown assembly language.
- UDSHMEM achieved optimal system-wide bandwidth, saturating system's capacity while respecting data layout in memory banks and data locality across physical nodes.
- UDSHMEM is part of the UpDown system presented in our IEEE TPDS 2026 article (TPDS'26).

Interweaving Project @ Northwestern University: Paths to OpenMP in the Kernel

- Achieved an average performance gain of 22% (geometric mean) across scales and benchmarks for runtime in kernel implementation by inspecting runtime behavior.
- Customized LLVM/OpenMP runtime library libomp and implemented pthread-embedded library to make libomp function within Nautilus kernel.
- Discovered a Floating-Point logic error in Nautilus codebase by benchmarking Gaussian elimination.
- Ported different benchmarks including NAS Parallel Benchmarks

Other Projects

Project Us @ MIT Media Lab: *Connecting Us*

- Led full-stack development of real-time cloud-based AI emotion recognition system with React, Flask, MongoDB and TensorFlow for initial client evaluation, pushing the project to participate in the MIT delta-v program.

AITom @ CMU: AI Tools for Tomography

- Led the research on 3D saliency detection for Cryo-ET by applying attention mechanism and teacher-student model in an unsupervised environment.

PUBLICATIONS

[arXiv'26] **W. Wang**, S. Shi, Y. Babuji, I. Foster, K. Chard, “**ExaServe: Large-Scale LLM Serving on Exascale HPC Systems**”, *preprint*, 2026 , arXiv:2609.10812.  

[TPDS'26] A. A. Chien, C. Colley, J. Ding, A. Fell, D. F. Gleich, H. Hoffmann, M. Jeje, R. Khandelwal, Y. Li, J. M. Monsalve-Diaz, M. Nourian, A. S. Rajasukumar, J. Su, T. Su, Y. Wang, **W. Wang**, R. Xu, T. Zhang, “**UpDown: A Supercomputer Co-Designed for Scalable Graph Processing**”, *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 2026 , Vol. 37, No. 7, pp. 1553–1570. 

[SC Posters'25] **W. Wang**, M. Gonthier, H. Lai, P. Nookala, H. Pan, I. Foster, I. Raicu, K. Chard, “**Exploring Fine-Grained Parallelism in Data-flow Runtime Systems on Many-Core Systems**”, *The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC25)*, *Research Poster*, 2025 . 

[SC Workshop'25] A. Fell, Y. Wang, T. Su, M. Nourian, **W. Wang**, J. M. Monsalve-Diaz, A. S. Rajasukumar, J. Su, R. Xu, R. Khandelwal, T. Zhang, D. Gleich, Y. Li, H. Hoffmann, A. A. Chien, “**KVMSR+UDWeave: Extreme-Scaling with Fine-grained Parallelism on the UpDown Graph Supercomputer**”, *The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC25)*, *Workshop*, 2025 . 

[IPDPS'25] **W. Wang**, M. Gonthier, P. Nookala, H. Pan, I. Foster, I. Raicu, K. Chard, “**Optimizing Fine-Grained Parallelism Through Dynamic Load Balancing on Multi-Socket Many-Core Systems**”, *2025 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, 2025 , Pages: 81-93. 

[arXiv'24] A. S. Rajasukumar, J. Su, Y. Wang, T. Su, M. Nourian, J. M. Monsalve-Diaz, T. Zhang, J. Ding, **W. Wang**, Z. Zhang, M. Jeje, H. Hoffmann, Y. Li, A. A. Chien, “**UpDown: Programmable fine-grained Events for Scalable Performance on Irregular Applications**”, *preprint*, 2024 . 

[LCPC'23] Y. Wang, A. S. Rajasukumar, T. Su, M. Nourian, J. M. Monsalve-Diaz, A. Pervaiz, J. Ding, C. Colley, **W. Wang**, Y. Li, D. F. Gleich, H. Hoffmann, A. A. Chien, “**Efficiently Exploiting Irregular Parallelism Using Keys at Scale**”, *Languages and Compilers for Parallel Computing (LCPC 2023)*, *Lecture Notes in Computer Science*, Springer, 2025 , pp. 78–95. 

[BDCAT'23] N. C. Hudson, J. G. Pauloski, M. Baughman, A. Kamatar, M. Sakarvadia, L. Ward, R. Chard, A. Bauer, M. Levental, **W. Wang**, W. Engler, O. P. Skelly, B. Blaiszik, R. Stevens, K. Chard, I. Foster, “**Trillion parameter ai serving infrastructure for scientific discovery: A survey and vision**”, *Proceedings of the IEEE/ACM 10th International Conference on Big Data Computing, Applications and Technologies*, 2023 . 

[SC'21] J. Ma, **W. Wang**, A. Nelson, M. Cuevas, B. Homerding, C. Liu, Z. Huang, S. Campanoni, K. Hale, P. Dinda, “**Paths to OpenMP in the Kernel**”, *International Conference for High Performance Computing, Networking, Storage and Analysis (SC21)*, 2021 . 

HONORS

- Intel Student Ambassador, 2025
- Crerar Fellowship, 2022
- Exceptional Funding of the Nation (China), awarded to the top 5%, the 12th National Innovation Training Program for College Students, 2018
- Gold Award, China College Students' Entrepreneurship Competition in Liaoning Province, 2018
- Nationwide Second Prize, China, “Innovation has a future” University AI Innovation Grand Competition, 2018

TEACHING

- MPCS 52040 Distributed Systems (Winter 2025, Autumn 2025, Spring 2026)

SKILLS

- Frameworks/libs: vLLM, PyTorch, OpenMP, Taskflow, SHMEM, MPI, Tensorflow, Flask, React, MongoDB.
- Programming Language: C, C++, Python, Java, Rust.